

What is software engineering?

Ideal Answer

Software engineering is a process of analyzing user requirements, designing, building, and testing software applications to satisfy these requirements.

Follow-up

What are the stages in software life cycle?

Q2

What is DevOps?

Ideal Answer

DevOps is a set of practices that combines software development and IT operations.

Follow-up

What is the aim of DevOps?

Expected Answer

To shorten the systems development life cycle and provide continuous delivery with high software quality.

Section 2 — Evolution vs Maintenance

Q3

What is the difference between software evolution and software maintenance?

Ideal Answer

Software maintenance means preserving software from failure or decline, while software evolution means continuous change from a lesser state to a better state.

Q4

Who distinguished between evolution and maintenance?

Ideal Answer

Lowell Jay Arthur.

Q5

According to Keith Bennett and Lie Xu, what is maintenance?

Ideal Answer

Maintenance is all post-delivery support.

Q6

What did Mehdi Jazayer say about software evolution?

Ideal Answer

What evolves is not the software but our knowledge about the software.

Section 3 — Lehman Laws

دي مهمة جدًا جدًا ومتوقع ييجي منها شفوي كثير.

Q7

Who pioneered software evolution research?

Ideal Answer

Lehman and his collaborators from IBM.

Q8

What are E-type systems?

Ideal Answer

Monolithic systems produced by a team within an organization that solve a real-world problem and have human users.

Q9

Explain Continuing Change law.

Ideal Answer

A system becomes less satisfying over time unless it is continually adapted to meet new needs.

Q10

Explain Increasing Complexity law.

Ideal Answer

A system becomes progressively more complex unless work is done to reduce the complexity.

Q11

What is Conservation of Familiarity?

Ideal Answer

The amount of new content in each release tends to stay constant or decrease over time.

Q12

What is Continuing Growth?

Ideal Answer

The functionality of a system increases over time to satisfy users.

Q13

What is Declining Quality?

Ideal Answer

A system loses quality over time unless its design is carefully maintained.

Section 4 — Software Maintenance**Q14**

Why is software maintenance needed?

Ideal Answer

Because there are always defects in delivered software and defect removal is not perfect.

Q15

What are the types of software maintenance according to Swanson?

Ideal Answer

Corrective, Adaptive, and Perfective maintenance.

Follow-up

What was added later?

Expected Answer

Preventive maintenance.

Q16

What is corrective maintenance?

Ideal Answer

Activities performed to eliminate discrepancies between expected and actual behavior.

Q17

What are enhancement activities?

Ideal Answer

Activities performed to implement changes in the system behavior or implementation.

Section 5 — SCM

دي برضو متوقعة جدًا.

Q18

What is SCM?

Ideal Answer

Software Configuration Management is the discipline of managing and controlling change in software evolution.

Q19

Why is SCM important?

Ideal Answer

To ensure released software is not contaminated by uncontrolled or unapproved changes.

Q20

What are the elements of SCM?

Ideal Answer

- Identification
 - Control
 - Auditing
 - Accounting status
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Section 6 — Reengineering

دي أكيد هنتيجي.

Q21

What is software reengineering?

Ideal Answer

It is the process of reorganizing and modifying existing software systems to make them more maintainable.

Q22

What is reverse engineering?

Ideal Answer

It defines a more abstract and easier-to-understand representation of the system.

Q23

Does reverse engineering change the software?

Ideal Answer

No, it examines the software without changing it.

Q24

What is forward engineering?

Ideal Answer

It is moving from high-level abstraction to physical implementation.

Q25

Write the reengineering formula.

Ideal Answer

Reengineering = Reverse Engineering + Forward Engineering + Change.

متوقعة جدًا Trick أسئلة

Trick 1

Does maintenance always mean evolution?

Answer

No.

Trick 2

Are Lehman laws applied to all systems?

Answer

No, mainly E-type systems.

Trick 3

Is reverse engineering a process of change?

Answer

No.

Trick 4

Which maintenance type prevents future problems?

Answer

Preventive maintenance.